

MIE377: Financial Optimization Models

Project 1 (Winter 2025)

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1 Introduction

The objective of this project was to develop an automated asset management system that achieves both optimal and consistent investment performance using mean-variance optimization (MVO). The model was trained, validated, and tested using provided data, which included asset prices and factor returns.

The final model was evaluated using two key metrics: the Out-of-Sample Sharpe ratio and the average turnover rate. The Sharpe ratio, measured over the entire investment horizon, followed the principle of "higher is better." The average turnover rate was assessed under the guideline that "lower is better." Additionally, the model was subject to a runtime constraint of five minutes.

Four trading algorithms were evaluated based on their performance against the specified criteria. Estimates of expected return (μ vector) and covariance (Q matrix) were refined using factor models. In addition to factor model estimates, μ and Q were directly estimated from asset returns and tested within the models. The final trading algorithm was selected based on the specified criteria.

Strengths, limitations, and other key insights of the final trading algorithm will be highlighted. Finally, a reflection on what could be done better in the future is detailed.

2 Methodology

This section will introduce the steps in our model selection process. The four MVO models used in testing, and their unique considerations, will be discussed. The regression models used to derive explanatory factors and estimates of μ and Q will be introduced, as well as their importance. Lastly, this section will provide details regarding the assessment of the MVO algorithms and regression models to provide an overview of the model selection process. Note that we allowed short-selling in all models.

2.1 MVO Algorithms

2.1.1 Model 1: Minimize Risk Optimization

Model 1 serves to minimize risk for a given return. The objective function minimizes portfolio variance subject to a target return and weight constraint. The structure of the model is as follows:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^T Q \mathbf{x} \\ \text{s.t.} \quad & \boldsymbol{\mu}^T \mathbf{x} \geq R, \\ & \mathbf{1}^T \mathbf{x} = 1. \end{aligned}$$

Model 1 is ideal for risk-averse individuals who seek to prioritize capital preservation but want to obtain a defined return, which was calculated by taking the average of all expected returns.

2.1.2 Model 2: Maximize Return Optimization

Model 2 serves to maximize return while imposing an upper bound on the allowed risk. The model maximizes the portfolio return subject to a risk limit and weight constraint. The model appears as follows:

$$\begin{aligned}
& \max_{\mathbf{x}} \quad \boldsymbol{\mu}^T \mathbf{x} \\
& \text{s.t.} \quad \mathbf{x}^T \mathbf{Q} \mathbf{x} \leq \epsilon^2 \\
& \quad \mathbf{1}^T \mathbf{x} = 1
\end{aligned}$$

Contrary to Model 1, this model is ideal for investors targeting high returns but allows for safety with a predefined risk limit. This limit must indicate a balanced portfolio as very low risk will result in low expected return, but very high risk will increase volatility, both of which decrease the Sharpe ratio. A risk limit was set to 0.06, a moderate risk level, for a balanced portfolio.

2.1.3 Model 3: Risk-Return Trade-Off Optimization

The risk-return trade-off optimization is a combination of Model 1 and Model 2 and serves to balance both risk and expected return. The model maximizes the difference in the portfolio return and risk, where the risk level is defined by a risk-aversion parameter. The model is also subject to a weight constraint, and is structured as follows:

$$\begin{aligned}
& \min_{\mathbf{x}} \quad \mathbf{x}^T \mathbf{Q} \mathbf{x} - \lambda \boldsymbol{\mu}^T \mathbf{x} \\
& \text{s.t.} \quad \mathbf{1}^T \mathbf{x} = 1
\end{aligned}$$

The choice of risk-aversion parameter differs by user needs. The selection of this parameter for this project is discussed more in *Section 2.2.4 Risk Aversion Parameters*.

2.1.4 Model 4: Robust Risk-Return Trade-Off Optimization with Ellipsoidal Uncertainty Set

Model 4 is built on Model 3 by introducing robustness to the algorithm. Robust risk-return optimization improves traditional optimization methods by addressing the issue of sensitivity to estimation errors in expected returns and the covariance matrix to make portfolios less affected by such inaccuracies. The model is as follows:

$$\begin{aligned}
& \min_{\mathbf{x}} \quad \lambda \mathbf{x}^T \mathbf{Q} \mathbf{x} - \left(\boldsymbol{\mu}^T \mathbf{x} - \epsilon \left\| \boldsymbol{\Theta}^{1/2} \mathbf{x} \right\|_2 \right) \\
& \text{s.t.} \quad \mathbf{1}^T \mathbf{x} = 1
\end{aligned}$$

A confidence level of 90% was used in the computation of epsilon, which indicates the size of the uncertainty set. A confidence level of 90% results in a larger estimation error range and allows for a balanced portfolio that manages both risk and return, which optimizes the Sharpe ratio. The testing that resulted in a confidence level of 90% is detailed in *3.5.1 Differing λ Data*.

2.2 Model Considerations

2.2.1 Model choices

Our motivation for utilizing the above models stem from the concepts learned in this course. We wanted to demonstrate our ability to successfully apply the algorithms using our practice from the lab sessions. Additionally, we wanted to make accurate and insightful conclusions to expand our knowledge of these models in a practical setting. In particular, we had an interest in robust optimization, as it is not widely used and would be a critical skill to have in the industry.

2.2.2 Short Selling

The goal is to maximize the portfolio Sharpe ratio while minimizing the average turnover rate, essentially crafting a theoretically optimal portfolio. The choice to allow or disallow short selling can have many implications that must be considered under the project framework. Disallowing short selling can decrease

the average turnover rate but limit the Sharpe ratio due to a lack of flexibility in constructing the optimal portfolio. Conversely, allowing short selling can better maximize the Sharpe ratio due to more flexibility in weight allocations, but may increase the average turnover rate. Considering the goal of this project, and the prioritization on maximizing the Sharpe ratio, all of the models listed in *2.1 MVO Algorithms* allow short selling.

2.2.3 Data Cleaning and Validation

In the calculated periodic returns of the assets, there were two invalid numbers in the matrix (NaN). This was resolved by setting the invalid values in the period returns matrix to zero to avoid an error in the MATLAB optimization solver, allowing it to run successfully.

Additionally, we chose to use all five years of data rather than subsetting it to avoid excessive overfitting. Reducing the dataset to less than five years would result in a model specific to the small dataset, affecting its ability to perform on new, unseen data.

2.2.4 Risk Aversion Parameters

Model 3 and Model 4 in *2.1 MVO Algorithms* require a user-inputted risk aversion parameter. This parameter defines how much weight is given to minimizing risk versus maximizing return to allocate weight to such assets accordingly. A baseline risk aversion parameter of 20 was used to test the models. A parameter of 20 is within the range for a balanced portfolio, providing a good measure of the Sharpe ratio and average turnover rate. This parameter was fine-tuned once the final model was selected in *3.5.1 Differing λ Data*.

2.3 Regularization Techniques and Factor Models

Regularization techniques were used to determine the factors that explain the most variation in the model. These best factors were used in forming a factor model that estimates μ and Q , which were used in calculating the Sharpe ratio and average turnover rate in the MVO models. Additionally, ubiquitous factor models can be used for the same purpose. It is crucial to use factor models to estimate μ and Q because they are typically more accurate and less noisy than direct estimates. Factor models also allow for better performance on new datasets as the estimates are more generalizable. In this section, the regularization techniques used to obtain these estimates, and the reasoning behind their forms, will be introduced, in addition to well-known factor models.

2.3.1 Best Subset Selection (BSS)

The penalized form of BSS uses the l_0 -norm, which is structured as follows:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_0$$

$$\|\mathbf{B}_i\|_0 = \mathbb{K}_{\alpha_i \neq 0} + \sum_{k=1}^p \mathbb{K}_{\beta_k \neq 0}$$

BSS selects the most important factors by setting a penalty on the number of non-zero coefficients which makes the norm values smaller. However, the penalized form does not allow explicit control on the number of non-zero elements. BSS promotes true sparsity, allowing several coefficients to be zero at optimality. However, the l_0 -norm is not convex, so BSS is not convex, resulting in BSS performing best on mixed-integer programming models with binary variables. Nevertheless, for completeness, insight, and curiosity, BSS was tested with the four algorithms listed previously.

2.3.2 Least Absolute Shrinkage and Selection Operator (LASSO)

The penalized form of LASSO uses the l_1 -norm and appears as follows:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_1$$
$$\|\mathbf{B}_i\|_1 = |\alpha_i| + \sum_{k=1}^p |\beta_{ik}|$$

LASSO penalizes the sum of absolute values of the factors which also makes the norm values smaller. Similar to BSS, LASSO promotes true sparsity resulting in zero-valued coefficients. The l_2 -norm is a convex function, so LASSO is also convex, which allows for efficiency when solving.

2.3.3 Ridge Regression (RR)

RR, the final regularization technique, utilizes the l_2 -norm. The penalized form is:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_2^2$$
$$\|\mathbf{B}_i\|_2 = \left((\alpha_i)^2 + \sum_{k=1}^p (\beta_{ik})^2 \right)^{1/2}$$

Uniquely, RR penalizes the Euclidean norm. However, RR does not promote true sparsity and instead attempts to shrink the coefficients to zero. However, the l_2 -norm is continuous, convex, and smooth, and allows a closed-form solution.

2.3.4 Fama-French Three Factor Model

This Three Factor Model uses the market factor, size factor, and value factor, and is given by the following:

$$r_i - r_f = \alpha_i + \beta_{im}(f_m - r_f) + \beta_{is}f_s + \beta_{iv}f_v + \varepsilon_i$$

Note that the factors are correlated in the Three-Factor Model as they are subsets of market data.

2.3.5 Capital Asset Pricing Model (CAPM)

The final factor model is CAPM which is a one-factor model of the market excess return. CAPM is known by the following:

$$r_i - r_f = \beta_{im}(f_m - r_f) + \varepsilon_i$$

2.3.6 Ordinary Least Squares Regression (OLS)

A simple OLS regression with the eight provided factors was also used. OLS regression minimizes the sum of squared residuals, similar to BSS, LASSO, and RR, but is subject to no constraints. OLS regression is given by:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2$$

2.4 Factor model considerations

The regularization techniques and factor models described above were chosen to estimate μ and Q for reasons similar to those in *2.2.1 Model Choices*. We wanted to apply a range of techniques learned in this course to develop a practical understanding of how each technique worked. This project served as an excellent opportunity for us to see the results from these methods, which expanded our understanding and built our intuition.

The regularization techniques can be used in their penalized or constrained forms. The LASSO and RR models are equivalent in their penalized or constrained form, but BSS is not. The penalized form of BSS was used because it is more flexible than its constrained form. The constrained form requires a manually selected cardinality constraint, which limits computational feasibility. Rather than arbitrarily estimating a cardinality constraint, the penalized form uses automatic factor selection which is better for large datasets where strict control is not necessary. It is better to adjust the penalty to make the norm smaller to obtain a balance between the number of factors and accuracy.

2.5 Assessment of Models

To properly evaluate the models, a baseline was established. This was accomplished by using the sample mean and covariance as direct estimates of μ and Q from the cleaned dataset to obtain Sharpe ratios and average turnover rates for each model, which allowed for a preliminary assessment of model performance. Furthermore, regression techniques and factor models were used to further refine the estimates of μ and Q . Sharpe ratios, average turnover rates, and runtimes were recorded for each factor model and MVO algorithm. More insightful observations were made regarding model performance. The model that achieved the largest Sharpe ratio and an acceptable average turnover rate was selected to be further optimized with parameter inputs, if applicable. The parameters were adjusted within a specified range and the optimized model was selected as best-performing based on its response to parameter changes, with close attention on the Sharpe ratio. General observations were made for each model, including a best-to-worst ranking.

3 Analysis

3.1 Baseline Data

The baseline data we began with for each model is provided in the table below.

Baseline	Model 1	Model 2	Model 3	Model 4
Sharpe Ratio	-0.001945	-0.04409	0.14652	0.16887
Average Turnover	0.073739	0.00000008	1140.35	0.85251
Runtime (s)	~ 2	~ 5	~ 1	~ 36

Table 1: Baseline data for models, with basic estimation of μ and Q from historic data.

Model 4 had the highest Sharpe ratio when using the baseline μ and Q values. While Model 3 had the second-highest Sharpe ratio, it exhibited an exceptionally high turnover rate. Therefore, Model 4 was the best initial choice.

We then began refining the optimization models by using OLS regression and factor models to estimate μ and Q . Tables 2-5 contain the baseline and refined results for all four optimization models.

3.2 Model 1 Data

Model 1	Baseline	OLS	CAPM	Three-Factor	BSS	LASSO	RR
Sharpe Ratio	-0.001945	0.069483	0.083962	0.07796	0.07678	0.081813	0.071452
Average Turnover	0.073739	0.2227	0.17484	0.18509	0.18383	0.18835	0.22596
Runtime (s)	~ 2	~ 1	~ 2	~ 1	~ 2	~ 2	~ 1

Table 2: Data collected for Model 1, Minimizing Risk Optimization.

All of the refined μ and Q estimates significantly improved the performance of Model 1. The CAPM factor model and the LASSO based factor model achieved the first and second highest Sharpe ratios, respectively. Their turnover rates were very similar, so the CAPM factor model was determined to be the best performing μ and Q estimator used in conjunction with Model 1.

3.3 Model 2 Data

Model 2	Baseline	OLS	CAPM	Three-Factor	BSS	LASSO	RR
Sharpe Ratio	-0.04409	0.052381	0.069067	0.063895	0.063026	0.068047	0.053804
Average Turnover	0.00000008	0.21969	0.1621	0.1738	0.17273	0.17465	0.22965
Runtime (s)	~ 5	~ 6	~ 7	~ 6	~ 7	~ 8	~ 6

Table 3: Data collected for Model 2, Maximizing Return Optimization.

All of the refined μ and Q estimates also significantly improved the performance of Model 2. The CAPM factor model and the LASSO-based factor model achieved the first and second highest Sharpe ratios, respectively. Their turnover rates were very similar, so the CAPM factor model was determined to be the best performing μ and Q estimator used in conjunction with Model 2.

3.4 Model 3 Data

Model 3	Baseline	OLS	CAPM	Three-Factor	BSS	LASSO	RR
Sharpe Ratio	0.14652	0.098093	0.099261	0.15429	0.13302	0.1222	0.1534
Average Turnover	1140.35	810.1	872.2	876.85	864.10	836.74	843.82
Runtime (s)	~ 1	~ 1	~ 1	~ 1	~ 1	~ 2	~ 1

Table 4: Data collected for Model 3, Risk-Return Trade-Off Optimization.

Model 3's Sharpe ratio only improved with estimates of μ and Q derived from the Fama-French three-factor model and ridge regression. The three-factor model yielded the highest Sharpe ratio, followed by ridge regression. However, all variations of Model 3 exhibited extremely high turnover rates, suggesting that its overall performance would not perform well against the assignment's evaluation criteria.

3.5 Model 4 Data

Model 4	Baseline	OLS	CAPM	Three-Factor	BSS	LASSO	RR
Sharpe Ratio	0.16887	0.10782	0.11163	0.11356	0.10288	0.083183	0.10463
Average Turnover	0.85251	1.45332	1.4961	1.4693	1.4577	1.6384	1.4472
Runtime (s)	~ 36	~ 38	~ 36	~ 28	~ 42	~ 69	~ 34

Table 5: Data collected for Model 4, Robust Risk-Return Trade-Off Optimization.

None of the factor model based estimations of μ and Q improved the performance on Model 4.

3.5.1 Differing λ Data

Since Model 4 achieved the highest Sharpe ratio with the baseline μ and Q estimation, we further optimized it by adjusting its risk aversion parameter, λ . We tested values from $\lambda = 15$ to $\lambda = 25$ (± 5 from $\lambda = 20$), and found λ peaked at $\lambda = 17$. Table 6 presents the results for Model 4 for $\lambda = 15$ to $\lambda = 20$.

Model 4, with lambda =	15	16	17	18	19	20
Sharpe Ratio	0.1715	0.17255	0.17272	0.1721	0.17079	0.16887
Average Turnover	1.2015	1.103	1.0228	0.95667	0.90046	0.85251
Runtime (s)	~38	~36	~28	~42	~69	~34

Table 6: Data for Model 4 with varying λ values.

The λ that achieved the highest Sharpe ratio was $\lambda = 17$. All λ 's had similar turnover rates, so $\lambda = 17$ was determined to be the optimal value of the risk aversion parameter. The confidence level, α , also demonstrated an increasing trend as it was gradually tested from 75% to 90%, where the Sharpe ratio peaked.

3.6 General Observations

Factor models did not always improve the performance of the optimization models (specifically Model 3 and Model 4), which was unexpected.

Since BSS typically performs best on mixed-integer programming models, it was expected to have subpar performance out of all the factor models. However, each factor model exhibited varying performance across different optimization models. As such, no specific factor model can be identified as the "best" or "worst". BSS did not perform particularly well, nor did it perform particularly poorly.

Since a model's Sharpe ratio accounted for 80% of its evaluation criteria, we prioritized maximizing the Sharpe ratio over minimizing turnover rates. Based on this approach, Model 4 was clearly the best-performing model, consistently achieving high Sharpe ratios across different factor models while maintaining acceptable turnover rates.

Although Model 3 consistently had the second-highest Sharpe ratios, its turnover rate was approximately 1,000 times higher than the other models, making it the worst-performing model. Models 1 and 2 had similar turnover rates, but Model 1 consistently achieved a higher Sharpe ratio across all factor model estimations. Therefore, Model 1 ranked as the second-best model, followed by Model 2 in third place.

4 Discussion

4.1 Strengths of Final Model

The robust risk-return trade-off optimization with ellipsoidal uncertainty had several key strengths compared to the other traditional MVO models analyzed. Unlike Models 1, 2, and 3, the ellipsoidal uncertainty set accounted for worst-case uncertainties in the asset return estimates. This reduced the risk of overfitting to historical data and made it more likely to perform well on unseen datasets. In addition, the ideal risk aversion parameter (λ), found to be 17, balanced the trade-off between minimizing risk and maximizing return effectively and optimized the portfolio's Sharpe ratio to 0.17272 for the given dataset.

Furthermore, traditional models like Model 1 and 2 are often unstable during volatile markets, so a robust optimization typically reduces turnover because it optimizes under worst-case conditions. This ensured a more reliable performance across risky markets and is evident in the model's low turnover value of 1.0228. Lastly, our final model did meet our expectations as it was built upon the first three models. Given what we learned in the course, we anticipated that a robust optimization model would yield the best results.

4.2 Limitations of Final Model

The final model relied on parameters like λ and α which had to be fine-tuned to produce an optimal portfolio. Since we were tasked with automating an asset management system, finding the best parameter values requires extensive back-testing which can be difficult to automate. This is because ideal values of λ and α vary across datasets and market conditions. Although the robustness of the model has major benefits, optimizing under worst-case conditions forces the model to assume that return estimates contain uncertainty and hedges against the worst case scenario, which can lead to over-conservatism in asset management. This means that the model may miss trading opportunities when the markets are less volatile.

4.3 Failure of Factor Model

After incorporating various factor model selection processes to provide estimates of μ and Q for the model's objective function, they failed to produce better results than the sample mean and covariance. Possible reasons for this include that factor models assume asset returns are driven by predefined risk factors which are theoretical and not guaranteed to explain actual price movements. Many hedge funds and asset managers mine their own factors rather than relying on standard theoretical models like Fama-French. In addition, the factors given are well known and mass incorporation of them over time reduces its effectiveness and are often arbitrated away.

Furthermore, factor models estimate expected returns and covariance using historical regressions, which may fail to capture changing risk dynamics in the data, leading to poor diversification in the portfolio. On the contrary, leveraging historical mean returns and sample covariance, like we did in our trading algorithm, provided a more reliable, data-driven approach that better reflected actual market conditions. This went against our expectations, as we had learned that factor models provided stronger estimates for μ and Q than using the sample mean and covariance estimates. However, in practice, their assumptions did not hold and surprisingly, the simple approach of using historical mean returns and sample covariance performed best.

5 Conclusion

This project successfully developed and evaluated an automated asset management system using robust risk-return trade-off optimization with an ellipsoidal uncertainty set. The model demonstrated superior performance in maximizing the Sharpe ratio while maintaining a relatively low turnover rate. By optimizing under worst-case conditions, the model effectively balanced risk and return, outperforming traditional MVO models, particularly in volatile markets.

Despite its strengths, the model's reliance on fine-tuned parameters such as the risk aversion parameter and the confidence level introduced challenges in automation, as these parameters require extensive back-testing and may vary with market conditions. Moreover, the assumption of incorporating uncertainty into return estimates, while advantageous in reducing risk, could lead to conservative decisions that miss trading opportunities during stable market conditions.

The integration of factor models, intended to refine estimates of expected return and covariance, did not yield the expected improvements. In fact, the use of historical mean returns and sample covariance emerged as the more reliable approach. This outcome suggests that while factor models are theoretically sound, they may not fully capture the dynamic nature of asset returns, leading to suboptimal diversification and performance in practical applications.

In future iterations, it is necessary to explore alternative approaches to factor modeling to understand why they yielded worse estimates of μ and Q than the sample mean and covariance estimates, enhancing the

robustness and accuracy of the system. Nonetheless, the findings from this project highlight the importance of careful model selection, validation, and parameter tuning when developing automated asset management systems.